

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination April/ May 2018

PH236/ PH223 Pharmaceutical Biochemistry

Date: 01.05.2018, Tuesday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The degradative processes are categorized under the heading of

[A] Anabolism
[B] Catabolism
[C] Metabolism
[D] None of the above

2. The digestive enzymes of cellular compounds are confined to

[A] Lysosomes
[B] Ribosomes
[C] Peroxisomes
[D] Polysomes

3. α -D-glucose and β -D-glucose are

[A] Stereoisomers
[B] Epimers
[C] Anomers
[D] Keto-aldo pairs

4. The intermediate in hexose monophosphate shunt is

[A] D-Ribulose
[B] D-Arabinose
[C] D-xylulose
[D] D-lyxose

5. The following is an enzyme required for glycolysis:

[A] Pyruvate kinase
[B] Pyruvate carboxylase
[C] Glucose-6-phosphatase
[D] Glycerokinase

6. Which one of the following is semi essential amino acid for humans?
- [A] Valine
 - [B] Arginine
 - [C] Lysine
 - [D] Tyrosine
7. A compound serving a link between citric acid cycle and urea cycle is
- [A] Malate
 - [B] Citrate
 - [C] Succinate
 - [D] Fumarate
8. Which of the following statement about the peptide bond is true?
- [A] It is a carbon-carbon bond
 - [B] It has cis hydrogen and oxygen atoms
 - [C] It is planar
 - [D] It has rotational freedom
9. An example of a saturated fatty acid is
- [A] Palmitic acid
 - [B] Oleic acid
 - [C] Linoleic acid
 - [D] Erucic acid
10. De novo synthesis of fatty acids requires all of the following except
- [A] Biotin
 - [B] NADH
 - [C] Pantothenic acid
 - [D] ATP
11. Cholesterol is the precursor for the biosynthesis of
- [A] Fatty acid
 - [B] Prostaglandins
 - [C] Bile acids
 - [D] Sphingomyelin
12. A Holoenzyme is
- [A] Functional unit
 - [B] Apo enzyme
 - [C] Coenzyme
 - [D] All of these
13. Isoenzymes are
- [A] Chemically, immunologically and electrophoretically different forms of an enzyme
 - [B] Different forms of an enzyme similar in all properties
 - [C] Catalysing different reactions
 - [D] Having the same quaternary structures like the enzymes
14. In competitive inhibition, the inhibitor
- [A] Competes with the enzyme
 - [B] Irreversibly binds with the enzyme
 - [C] Binds with the substrate

- [D] Competes with the substrate
15. 2, 4-Dioxypyrimidine is
- [A] Thymine
[B] Cytosine
[C] Uracil
[D] Guanine
16. The key substance in the synthesis of purine, phosphoribosyl pyrophosphate is formed by
- [A] α -D-ribose 5-phosphate
[B] 5-phospho β -D-ribosylamine
[C] D-ribose
[D] Deoxyribose
17. In the purine nucleus, carbon 6 is contributed by
- [A] Glycine
[B] CO_2
[C] Aspartate
[D] Glutamine
18. Haemoproteins that generate ATP via electron transport chain are called as
- [A] Flavoproteins
[B] Cytochrome
[C] Haemoglobin
[D] None of above
19. Enzyme that makes ATP by chemiosmosis is
- [A] ATP dehydrogenase
[B] Gyrase
[C] ATP synthase
[D] Dehydrogenase
20. Oxidative phosphorylation occurs in
- [A] Mitochondria
[B] Nucleus
[C] Cell membrane
[D] Ribosomes

SECTION – II

- Q 2** Attempt any **TWO** of the following:
Write in brief biological role of following molecules:
- | | | | | |
|----------|--------------------------|---------------|------------------|-----------|
| A | I. ATP | II. Ubiquinon | III. Cytochrome | 05 |
| | IV. Iron sulphur protein | | V. Flavoproteins | |
- B** What do you mean by allosteric regulation of enzyme activity? With any one metabolic reaction, explain negative feedback regulation. **05**
- C** Discuss the types of chemical bonds present in the nucleic acids and their role in the formation and stability of DNA. **05**
- Q 3** Attempt any **TWO** of the following:
- A** With suitable diagram, explain chemiosmotic theory of oxidative phosphorylation. **05**

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|----------|---|-----------|
| B | With the graphical representations, derive and explain Michaelis-Menten constant. | 05 |
| C | With a suitable pathway, describe biosynthesis of ketone bodies. | 05 |

SECTION – III

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| Q 4 | Attempt any <u>FOUR</u> of the following: | |
| A | Write a note on TCA Cycle. | 05 |
| B | Define following terminologies:
i) Gluconeogenesis ii) Glycogenolysis iii) Mutarotation iv) Glycogenesis
v) Apoenzymes | 05 |
| C | Explain structure and working principle for ATP synthase complex. | 05 |
| D | Add a note on β - oxidation of even carbon numbered fatty acids. | 05 |
| E | What are Amino Acids? Classify them structurally with suitable examples. | 05 |
| F | How Inosine Monophosphate is synthesized in our body? Write down the respective pathway. | 05 |
| Q 5 | Attempt any <u>FOUR</u> of the following: | |
| A | Explain a pathway for biosynthesis of urea. | 05 |
| B | Define various orders of protein structures and describe β - sheets. | 05 |
| C | Write a pathway for biosynthesis of cholesterol. | 05 |
| D | What are carbohydrates? Describe reactions of Monosaccharide. | 05 |
| E | Write down a structure of Purine and Pyrimidine nucleus. Show source of different atoms in their structure. | 05 |
| F | Name various types of macromolecules present in our body and write down biological role of each of them. | 05 |
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